Topic 7: Plants – Structure, photosynthesis and crop yield

Students should:		Guidance *iLowerSecondary Curriculum reference*
7.1	know the functions of roots, stems and leaves	know that roots anchor the plant and absorb water and minerals minerals to include nitrates for growth and magnesium ions for chlorophyll root hairs increase the surface area of the root stems support the plant and transport materials throughout the plant leaves are where photosynthesis occurs *Year 7: Plants – External structure of plants*
7.2	know the simple external features of plants living in different habitats	for example, compare leaf structure, leaf size, leaf texture and closeness of leaves in desert, arctic and temperate zone plants *Year 7: Plants – External structure of plants*
7.3	know how water and minerals are absorbed and transported in flowering plants	know that water and dissolved minerals are absorbed by the roots and transported up the stem through xylem vessels *Year 8: Plants – Transport of water and minerals*
7.4	explain how fertilisers can increase crop yield	plan/interpret simple investigations into the effect of fertiliser on plant growth understand how to interpret data from such investigations, presented as tables or graphs *Year 8: Plants – Fertilisers*
7.5	understand how the structure of the leaf is adapted for photosynthesis	know that leaves have a large surface area to capture sunlight, a waxy upper surface to prevent water loss and stomata to allow carbon dioxide into the leaf and oxygen out of the leaf use the terms *chlorophyll*, *chloroplasts* and *photosynthesis* correctly *Year 9: Plants – Photosynthesis and crop yield*
7.6	know how to model photosynthesis using a word equation and a balanced symbol equation	*Year 9: Plants – Photosynthesis and crop yield*

Students should:		Guidance *iLowerSecondary Curriculum reference*
7.7	explain how light intensity and carbon dioxide concentration affect the rate of photosynthesis	describe/interpret simple experimental procedures and results from investigating the effects of light, carbon dioxide and temperature on the rate of photosynthesis *Year 9: Plants – Photosynthesis and crop yield*
7.8	explain how crop yield may be affected by changes in abiotic factors	abiotic variables include rain, wind, temperature, soil, pollution, nutrients, pH, and sunlight *Year 9: Plants – Photosynthesis and crop yield*
7.9	know how selective breeding can lead to new plant varieties	understand the stages of selective breeding *Year 9: Plants – Photosynthesis and crop yield*